week 4 Stanford CS3N1 Regularization/Lossfunctions

x → Wi+b = Logit bic of fashion MNST 10 classes
[x, y, z, v, b, n]
biggest logit = probability

logit determines prediction
Loss function is how good prediction is → backpropogation → gradients → update weights

Different feotures different scales have different predictions bic of numerical scale
Zero-centering
[5, 7, 9] mean = 7 Subtract mean = 0 from
[-2, 0, 2]
Three data sets
Traning, validation, test
↳ only do traning prevent data Leakage
PCA: Find the direction in which data has most variation
Variance: How much one variable vary
Covariance: How two variables vary
covariance matrix
↳ Which space has most variation
↳ Egnivectors tells how much variation happens along the directions
Low variance components remove tiny details but finer information disappears
Whitening tries to make variance same in every direction
Randomly generate weight
Bath Norm calculation btween activation layers
Regularization: Minimize error, but also no complicated solution
Dropout: When overfitting when do ya stop
classification = categories
Regression = predict numbers